CGPA CALCULATOR USING PYTHON DJANGO

MINI PROJECT REPORT

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CERTIFICATE

This is to certify that S. Anuradha (23AK1A0411), Department of ECE, has successfully completed the mini project titled 'CGPA Calculator Using Python Django' as part of the academic curriculum.

ABSTRACT

The CGPA Calculator is a web application developed using Python Django. The system accepts credits and grades as input, calculates CGPA automatically, and displays the result. The application is simple, accurate, and user friendly.

OBJECTIVES

1. Automate CGPA calculation.
2. Reduce manual errors.
3. Provide accurate results.
4. Learn Django framework.

SOFTWARE REQUIREMENTS

Python 3.x
Django Framework
VS Code
HTML
Web Browser

WORKING

The user enters credit and grade values. Django receives the form data, calculates CGPA using the formula and displays the result page.

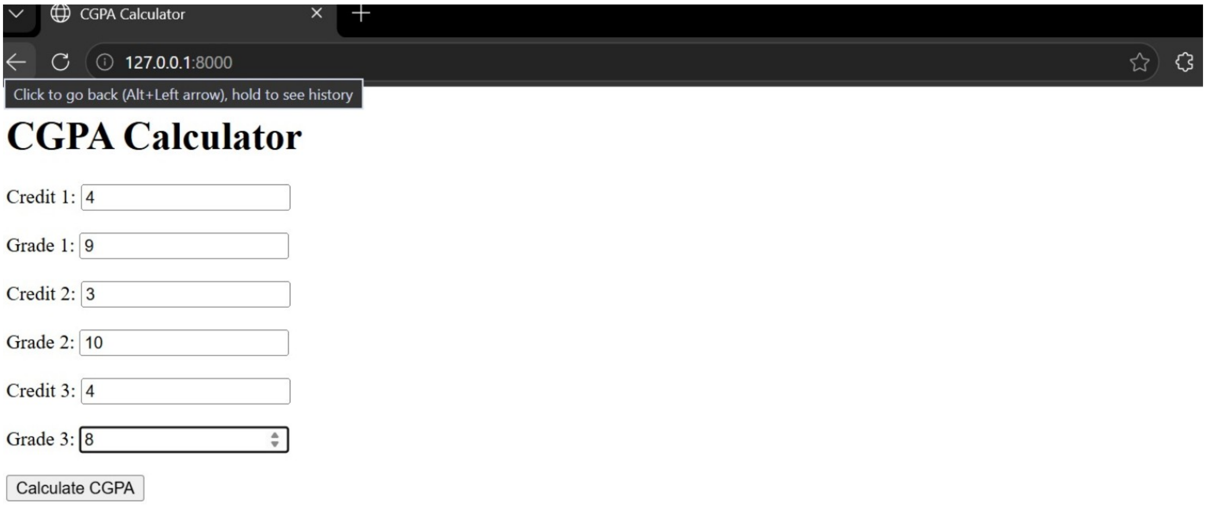
SOURCE CODE - views.py

```
from django.shortcuts import render

def home(request):
  return render(request,'calculator/home.html')

def calculate(request):
  c1=int(request.POST['c1'])
  g1=int(request.POST['g1'])
  c2=int(request.POST['c2'])
  g2=int(request.POST['g2'])
  c3=int(request.POST['c3'])
  g3=int(request.POST['g3'])
  cgpa=round(((c1*g1)+(c2*g2)+(c3*g3))/(c1+c2+c3),2)
  return render(request,'calculator/result.html',{'cgpa':cgpa})
```

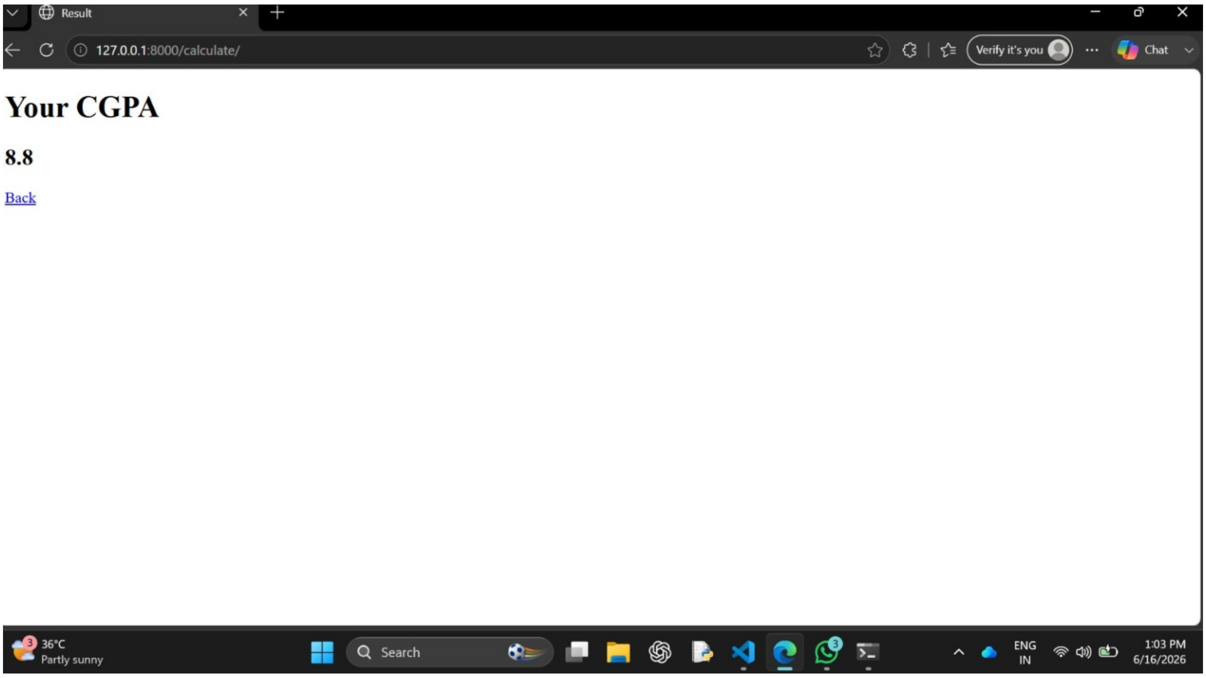
SOURCE CODE - urls.py

```
from django.urls import path
from . import views

urlpatterns=[
  path('',views.home),
  path('calculate/',views.calculate),
]
```

SCREENSHOT - HOME PAGE



SCREENSHOT - RESULT PAGE



RESULT AND CONCLUSION

The project was successfully implemented using Python Django. For the entered values (Credits: 4,3,4 and Grades: 9,10,8), the calculated CGPA is 8.8. The application works correctly and provides accurate results.

REFERENCES

1. Python Documentation
2. Django Documentation
3. HTML Documentation